

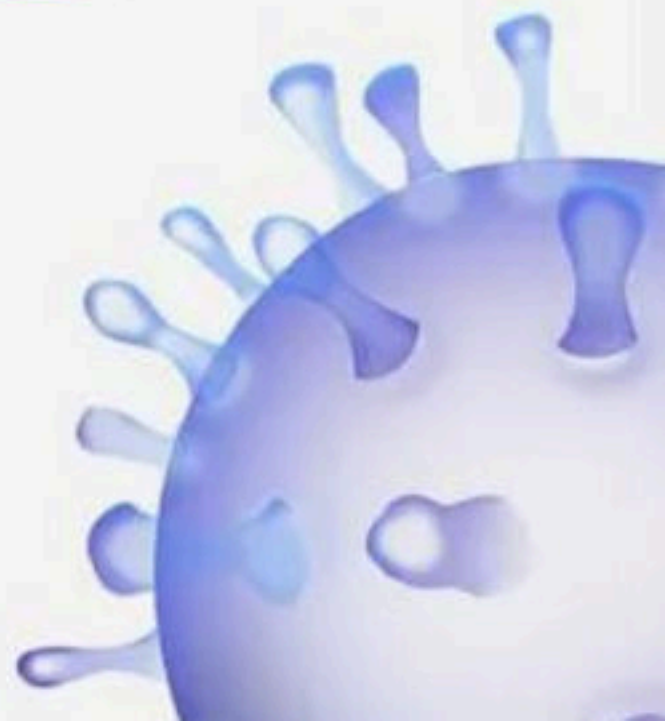
Organism and Population - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



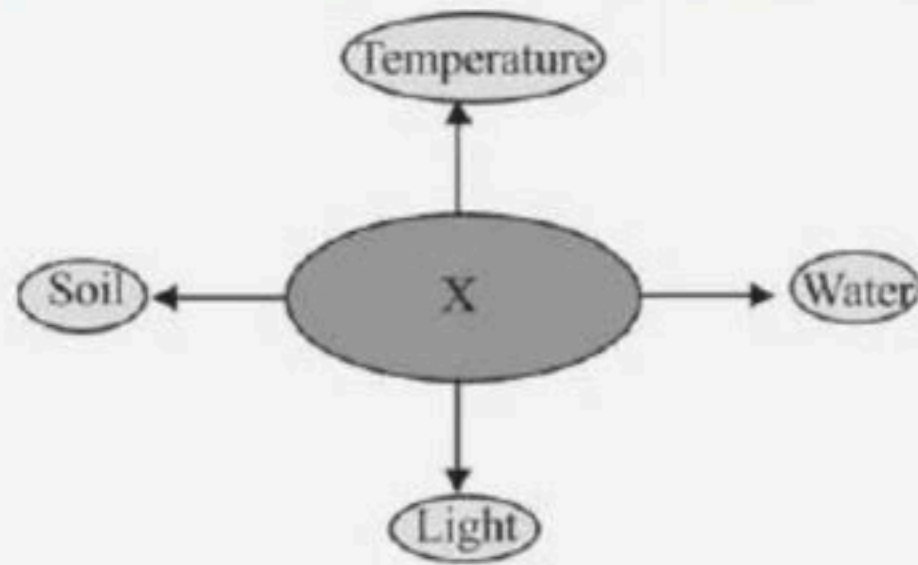
Good evening
↓
Ready ↓

Organism and Population ✓ - 100 MCQs



BIOLOGYLIVE

1. What does 'X' denote in the following diagram.



- (1) Key abiotic process
- (2) Key biotic process
- (3) Key biotic factor
- (4) Key abiotic factor



② ^{V. Imp} In the following expression $\underline{N_t} = \underline{N_0} + \underline{B} + \underline{I} - D - E$, where

I, B and D stand for :

- (1) Emigration rate, natality rate, mortality rate
- (2) Mortality rate, natality rate, immigration rate
- (3) Mortality rate, immigration rate, natality rate
- ✓ (4) Immigration rate, natality rate, mortality rate



3. The interaction in which one species is benefited only is known as

(1) Parasitism ✓

(2) Predation ✓

✓ (3) Both 1 and 2

(4) Mutualism



4. Habitat includes :-

- (1) Biotic components ✓
- (2) Abiotic components ✓
- (3) Pathogens, predators, parasites and competitors ✓
- (4) ✓ All the above



5. Barnacles growing on the back of a whale are

- ✓✓ (1) Marine crustacean
- (2) Fresh water crustacean
- (3) Marine insects
- (4) Marine arachanida



6. The age pyramid with broad base indicates

- ✓ (1) High percentage of pre reproductive people
- (2) Low percentage of reproductive people
- (3) High percentage of old individuals
- (4) None of the above



7. Mutualism is an interaction between two species in which



- ✓ (1) Both are benefited
- (2) Both are harmed
- (3) One is benefited and the other is harmed
- (4) One is benefited and the other is neither benefited nor harmed



8. Identify the incorrect pair regarding populations interactions

I. Human- Lice ✓

II. Dogs- Ticks ✓

III. Cuscuta- Hedge plant ✓

IV. Marine fishes - Copepods ✓

(1) I, II

(2) II, III

(3) III only

✓ (4) None



9. **Assertion** : The collective and continuous growth of animals and plants in an ecological area is called biome

Reason : Coral reefs have lowest biomes 

(1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.

 (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.



10. Snails and fishes undergo :- Aestivation

(1) Hibernation

(2) Diapause

(3) Migration

✓ (4) Aestivation = Summer Sleep



11. What percentage of organisms are conformers,
respectively? v. imp

- (1) 90 percent of animals and nearly all plants
- (2) 1 percent of animals and nearly all plants
- (3) 100 percent of animals and nearly all plants
- ✓ (4) 99 percent of animals and nearly all plants



12. Which of the following statements is not true about parasitism?

- (1) The host is generally larger than the parasite ✓✓
- ✓ (2) An efficient parasite has developed features to kill its host ✓
- (3) Smaller parasites often live as endoparasites within the body of the host
- (4) Ectoparasites are attached to the outside of the host's body by specialized organs



13. The gradual physiological adjustment to slowly changing new environmental conditions is known as

- (1) Selection
- (2) Introduction
- ✓ (3) Acclimatization
- (4) Quarantine



14. The integral form of the exponential growth equation is as follows

(1) $N_t = N_0 e^{RT}$

✓ (2) $N_t = N_0 e^{rt}$

(3) $N_t = N_0 e^{rt}$

(4) $N_t = N_0 e_{rt}$



15. If a population grows in a habitat with limited resources, then following phases of achievement will be observed-

→ Sigmoid

(1) Lag → Asymptote → Deceleration → Acceleration

(2) Lag → Deceleration → Acceleration → Asymptote

✓ (3) Lag → Acceleration → Deceleration → Asymptote

(4) Lag → Acceleration → Asymptote → Deceleration



16. Demography is the study of :-

- ✓✓ (1) Population growth ↪
- (2) Birth rates
- (3) Family pedigrees
- (4) Death rates



17. Which one of the following population interactions is widely used in medical science for the production of antibiotics?

- (1) Parasitism
- (2) Mutualism
- (3) Commensalism
- ✓ (4) Amensalism



18. Mycorrhiza is an example of

(1) Endoparasitism

(2) Decomposers

✓ (3) Symbiotic relationship

(4) Ectoparasitism



**19. Biological control methods in agriculture pest control
is mainly based on which biological interaction**

(1) Commensalism

(2) Parasitism

✓ (3) Predation

(4) Mutualism



20. All the following interactions are mutualism, except:

- (1) Association of algae and fungi in lichens
- (2) Association of fungi and roots of higher plants in mycorrhiza
- (3) Plant and animal relation for pollination
- ✓ (4) Association of cattle egret and grazing cattle = commensalism



21. Adaptations that have evolved over a long evolutionary time and are Genetically. fixed.

- ✓ (1) Genetically
- (2) Phenotypically
- (3) Mutationally
- (4) Randomly



22 Match the following.

Column I		Column II	
A	Predation	P	<i>Cuscuta</i> and hedge plants
B	Commensalism	Q	<i>Balanus</i> and <i>Chthamalus</i>
C	Parasitism	R	<u>Cactus</u> and <u>moth</u>
D	Competition	S	Orchid and mango

- ☒ (1) A – R, B – S, C – P, D – Q
(2) A – S, B – R, C – Q, D – P
(3) A – P, B – R, C – Q, D – S
(4) A – R, B – S, C – Q, D – P



23. Salt concentration (Salinity) of the ^{" "}sea measured in parts per thousand is:

(1) 10-15

(2) 30-70

(3) 0-5

☒ (4) 30-35



24. Which of the following characteristics of the soil determine the percolation and water holding capacity of the soils such as ∴

- (1) Soil composition only
- (2) Grain size and aggregation alone
- ✓ (3) ✓ Soil composition, ✓ grain size and aggregation
- (4) Soil composition, aggregation alone



25. Evolutionary biologist believe that the success of mammals is largely due to their ability to maintain a

- (1) Constant body temperature ✓✓
- (2) Constant osmotic concentration ✓✓
- ✓✓ (3) Both (1) and (2) ✓✓
- (4) None of these



26. Which of the following would necessarily decrease the
density of a population in a given habitat?

- (1) Natality > mortality
- (2) Immigration > emigration
- ✓ (3) Mortality and emigration
- (4) Natality and immigration



27. Which one is the correct statements for logistic model of population growth? 1-1mp

I. As population increases the competition goes on increasing. ✓

II. All individuals have same effect on population growth. ✗

III. There are limited natural resources. ✓✓

IV. Population growth rate increases as the size of population approaches the carrying capacity. ✗

(1) I and II

(2) Only IV

(3) IV and III

✓ (4) I and III



28. Which of the following is a population attribute?

- ✓ (1) Majority of them live in groups in a well-defined geographical location
- (2) Never share or compete for similar resources
- (3) Potentially never interbreed
- (4) Single individuals of any species cannot live in isolation



29. Select the correct answer :-

I. Many marine fish are infested with endoparasitic copepods. ✗

II. The female mosquito is not considered a parasite,
although it needs our blood for reproduction. ✓

(1) Only I is correct

✓ (2) Only II is correct

(3) Both are correct

(4) None are correct



30. Which of the following is not an example of commensalism?

- (1) Sea anemone and clown fish
- (2) Orchid on mango branch
- ✓ (3) Tapeworm and man → endoparasitism.
- (4) Cattle egret and grazing cattle



31. Population is best described by organisms that

- (1) Potentially interbreed ✓
- (2) Share or compete for similar resources ✓
- (3) Live in groups in a well defined geographical area ✓
- (4) ✓ All of them



32. The human population first began to grow
exponentially at the time of

(1) Tool using revolution

✓ (2) Agricultural revolution ✓

(3) Industrial revolution

(4) First world war



33. Organisms which can tolerate slightest variation in salt concentration are

↓
Short

- ✓ (1) Stenohaline
- (2) Stenophagic
- (3) Stenothermal
- (4) Stenohydric



34 Microbes present in hydrothermal vents where the temperature far exceed 100°C is :-

(1) Cyanobacteria

✓(2) Archaeobacteria

(3) Eubacteria

(4) None of these



35. In the following reaction,

$$dN/dt = (b - d) \times N$$

Substituting r in place of b-d then

$$dN/dt = rN$$

Here 'r' is known as

- ☒ (1) Intrinsic rate of natural increase
- (2) Intrinsic rate of natural decrease
- (3) Extrinsic rate of natural decrease
- (4) Extrinsic rate of natural increase



36. Thermoregulation is energetically expensive to ____

✓ (1) Shrews and humming birds ↩

(2) Elephants and mouse

(3) Kangaroo rats

(4) Desert lizard



37. Match Column - I with Column - II

Choose the correct answer from the options given below.

Column I

A Allen's Rule

B Physiological adaptation

C Behavioural adaptation

D Biochemical adaptation

(1) A - S; B - Q; C - R; D - P

✓ (3) A - S; B - P; C - Q; D - R

Column II

P Kangaroo rat

Q Desert lizard

R Marine fish at depth

S Polar seal

(2) A - P; B - S; C - R; D - Q

(4) A - S; B - R; C - Q; D - P



38. Match the columns I and II with respect to measured as salinity in parts per thousand

Column I		Column II	
A	Inland waters	P	> 100
B	Sea	Q	Less than 5
C	Hypersaline-laggon	R	30-35

2025

- (1) A – R, B – P, C – Q
✓ (2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



39. Match Column I (Interacting species) with Column II (Name of interaction) :
Choose the correct answer from the options given below:

Column I		Column II	
A	A Leopard and a Lion in a forest/grassland →	P	Competition
B	A Cuckoo laying <u>egg</u> in a Crow's nest →	Q	Brood parasitism
C	Fungi and root of a higher plant in Mycorrhizae →	R	Mutualism
D	A cattle egret and a Cattle in a field →	S	Commensalism

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- ✓ (1) A-P, B-Q, C-R, D-S
(2) A-P, B-Q, C-S, D-R
(3) A-R, B-S, C-P, D-Q
(4) A-Q, B-R, C-P, D-S



40. Match Column-I with Column-II and choose the correct option.

Column I		Column II	
A	Pacific salmon fish	P	Verhulst - pearl logistic growth
B	$N_t = N_0 e^{rt}$	Q	Breed only once in <u>life</u> time
C	Oyster	R	Exponential growth
D	$\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$	S	A large number of small sized offsprings

- (1) A – S, B – R, C – P, D – Q
(2) A – R, B – S, C – P, D – Q
(3) A – R, B – P, C – S, D – Q
(4) A – Q, B – R, C – S, D – P



41. Match Column I (Interaction) with Column II (Species A and B) :
Choose the correct answer from the options given below:

Column I		Column II	
A	Mutualism	P	$+(A), 0(B)$
B	Commensalism	Q	$-(A), 0(B)$
C	Amensalism	R	$+(A), -(B)$
D	Parasitism	S	$+(A), +(B)$

(1) A-S, B-Q, C-P, D-R

✓ (2) A-S, B-P, C-Q, D-R

(3) A-S, B-R, C-P, D-Q

(4) A-R, B-P, C-S, D-Q



42. Match the columns I and II.

Column I

A Morphological

B Physiological adaptation

C Chemical substance

D Behavioural adaptation

Column II

P Altitude sickness

Q Cactus

R Caffeine

S Basking of lizards

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

☒ (4) A – Q, B – P, C – R, D – S



43. Match the columns I and II.

Column I

A Hibernation

B Aestivation

C Diapause

Column II

P Snails and fish

Q Zooplankton

R Bears

✓ (1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



44. Match the columns I and II.

Column I

A Regulate

B Conform

C Migrate

D Suspend

Column II

P Siberian birds

Q Bacteria

R All plants

S Humans



(1) A – R, B – P, C – Q, D – S

☒ (2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



45. Match the columns I and II.

Column I		Column II	
A	North American deserts	P	CAM
B	Opuntia	Q	Internal <u>fat oxidation</u>
C	Desert lizards	R	Bask in the sun
D	Allen's rule	S	Mammals from colder climates

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

✓ (4) A – Q, B – P, C – R, D – S



46. Match the columns I and II.

Column I

A Growing

B Stable

C Declining

Column II

P Triangular

Q Urn

R Bell

} Sharp, al
pyramid

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

☒ (3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



47. Match the columns I and II.

Column I

A Mutualism

B Commensalism

C Parasitism

D Predation

Column II

P Mosquito

Q Tiger and deer

R Cran and sea anemone

S Fig and wasp

(1) A – R, B – P, C – Q, D – S

✓ (2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



48. Match the columns I and II.

Column I		Column II	
A	Pacific salmon fish	P	Breed only once
B	Most birds and mammals	Q	Small-sized offspring
C	Oysters	R	Breed many times

(1) A – R, B – P, C – Q,

(2) A – Q, B – R, C – P,

✓ (3) A – P, B – R, C – Q,

(4) A – Q, B – P, C – R



49. Match the columns I and II with respect to the

Column I		Column II	
A	Temperate forest	P	-11°C to 0°C
B	Arctic and Alpine Tundra	Q	10 to 20°C
C	Grassland	R	0 to 25°C

- (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



50. Relate Column A with Column B with regard to predatory behaviour

Column I		Column II	
A	<i>Calotropis</i>	P	Invertebrates
B	<i>Pisaster</i>	Q	Distasteful
C	Monarch butterfly	R	Cryptically coloured
D	Frogs	S	<u>Cardioglycoside</u>

(1) A – S, B – P, C – R, D – Q

✓ (2) A – S, B – P, C – Q, D – R

(3) A – Q, B – S, C – P, D – R

(4) A – S, B – P, C – R, D – Q



51. Match Column I with Column II:

Choose the correct answer from the options given below:

Column I		Column II	
A	Logistic growth	P	Unlimited resource availability condition
B	Exponential growth	Q	Limited resource availability condition
C	Expanding age pyramid	R	The percent <u>individuals</u> of pre-reproductive age is <u>largest</u> followed by reproductive and post reproductive age groups
D	Stable age pyramid	S	The percent individuals of <u>pre-reproductives</u> and <u>reproductive</u> age group are <u>same</u>

✓(1) A-Q, B-P, C-R, D-S

(2) A-Q, B-R, C-P, D-S

(3) A-Q, B-S, C-P, D-R

(4) A-Q, B-S, C-R, D-P



52. Match the type of adaptation given in Column-I with their examples given in Column-II. Select the option showing correct combination.

Choose the correct option from the following:

Column I		Column II	
A	Biochemical adaptation	P	Desert lizards
B	Behavioural adaptation	Q	Deep sea fishes
C	Physiological adaptation	R	<i>Opuntia</i>
D	Morphological adaptation	S	Kangaroo rats

(1) A – Q, B – R, C – S, D – P

(2) A – P, B – Q, C – R, D – S

☒ (3) A – Q, B – P, C – S, D – R

(4) A – S, B – R, C – Q, D – P



53. Match the columns I and II.

Column I		Column II	
A	Natality →	P	Number of births during a given period
B	Mortality	Q	Number of individuals of the same species that have come into the habitat
C	Immigration	R	Number of <u>deaths</u> during a given period
D	Emigration →	S	Number of individuals of the population who left the habitat

(1) A – R, B – P, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

✓ (3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P, C – R, D – S



54. Statement A : Predation is an interspecific interaction with a feeding strategy. ✓✓

Statement B : Predators help in maintaining species diversity in a community

(1) Statement A is correct but Statement B is incorrect.

(2) Statement A is incorrect but Statement B is correct.

✓ (3) Both statements are correct.

(4) Both Statements are incorrect.



55) Select the not false statement

I. Some organisms are able to maintain homeostasis by physiological (sometimes behavioural also) means which ensure constant body temperature, constant osmotic concentration etc. are called Regulators.

II. Snow leopards are found in Kerala forests and tuna fish are rarely caught beyond tropical latitudes in the ocean. ✗

III. Water is the least important factor influencing the life of organisms ✗

✓ (1) I is correct

(2) II is correct

(3) III is correct

(4) I, II, III are correct



56. Statement A : The eggs of the parasitic bird have evolved to resemble the host's egg in size and colour to reduce the chances of the host bird detecting the foreign eggs and ejecting them from the nest. ✓✓

Statement B : Koel is the host and crow is the parasite ✗

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



57. Consider the following statements. ∴ -

I. Brood parasitism in birds is an example of parasitism in which the parasitic bird lays its eggs in the nest of its host and host incubates them.

II. During the course of evolution, the eggs of the parasite bird have evolved to resemble the host's eggs in size and colour and reduce the chances of the host bird detecting the foreign eggs and removing them from the nest.

(1) I is true, but II is false

(2) II is true, but I is false

☒ (3) Both I and II are true

(4) Both I and II are false



58 Statement A : Many desert plants have Sunken stoma to minimise water loss. ✓

leaves
Statement B : In Opuntia ~~stem~~ is modified (or) reduced to spines ✗

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



59. Given below are two statements:

Statement A : Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot coexist indefinitely and competitively inferior one will be eliminated eventually.

Statement B : In general, carnivores are more adversely affected by competition than herbivores. In the light of the above statements, choose the correct answer from the options given below: ✓

- (1) Both Statement A and Statement B are true.
- (2) Both Statement A and Statement B are false.
- ✓ (3) Statement A is correct but Statement B is false.
- (4) Statement A is incorrect but Statement B is true.



60. Statement A : Nearly 25 per cent of all insects are known to be phytophagous.

Statement B : The Monarch butterfly is highly tasteful to its predator

- ☒ (1) Statement A is correct but Statement B is incorrect. ✗
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



61. Read the statements and identify the correct option.

XI. Percentage of Phytophagous insects are nearly 45%.

II. Many plants produce and store chemicals that make the herbivore sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it. ✓

III. The Monarch butterfly is highly distasteful to its predator. ✓

(1) I, II are wrong

✓ (2) I is wrong

(3) All are wrong

(4) All are correct



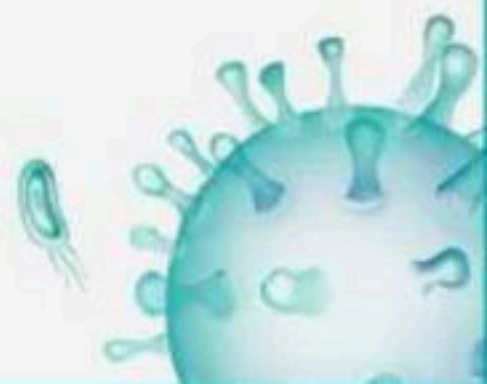
62. In Connell's elegant field experiments the superior barnacle ^{Balanus} _____ dominates the intertidal area, and exclude smaller barnacle _____.

(1) Chathamalus, Balanus

✓ (2) Balanus, Chathamalus

(3) Chathamalus, Chhathamalus

(4) Balanus, Balanus



63. In some shallow South American lakes, which of the following compete for the same food of lake?

(1) Visiting flamingoes ✓

(2) Resident fishes ✓

✓ (3) Both (1) and (2)

(4) None of these



64. Which of the following is ["]not a ["]parasitic adaptation?

- (1) Loss of unnecessary sense organs
- ✓ (2) Absence of adhesive organs or suckers
- (3) Loss of digestive system
- (4) High reproductive capacity



65. Following are correct example of ectoparasite

- (1) Lice on humans
- (2) Ticks on dogs
- (3) Copepods on marine fishes
- ☒ (4) All are correct



66. While explaining interspecific interaction of population, sign is assigned for beneficial interaction, (-) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned for one species and (-) for another species involved in the interaction?

- (1) Amensalism
- (2) Commensalism
- (3) Competition
- (4) Predation



67. Animals obtaining nourishment from other animals by saprozoic nutrition are called

(1) Mesophytes

✓ (2) Parasites ✓

(3) Xerophytes

(4) Halophytes



68. If a new habitat is just being colonised, ____ X ____ ^{Immig.} may contribute more significantly to population growth than ____ Y ____

- ✓ (1) X - Immigration, Y - Birth rates
- (2) X - Emigration, Y - Birth rates
- (3) X - Immigration, Y - Death rates
- (4) X - Emigration, Y - Death rates



69. Which one of the following is a total plant parasite?

(1) Copepods

(2) Calotropis

✓ (3) Cuscuta

(4) Lichens



70. Poisonous cardiac glycosides are found in

- (1) Acacia plant
- (2) Sea anaemone
- ✓(3) Calotropis plant
- (4) Lantana Plant



71 Best example of co-evolution

- ✓ (1) Fig trees and pollinator species of wasp
- (2) Fig trees and pollinator species of moth
- (3) Sea anemone and clown fish
- (4) Orchids and bees



72. Which of the following is correct in a predator-prey cycle?

- (1) A decline in the numbers of prey causes a increase in the number of predators
- (2) A decrease in the numbers of prey causes a decrease in the number of predators
- (3) An increase in the number of predators triggers an increase in the numbers of prey
- (4) A decrease in the numbers of predators causes a decrease in the number of prey



73. The life cycles of endoparasites are more complex because of their extreme specialisation. They are

(1) Morphological and anatomical features are greatly simplified ✓

(2) High reproductive potential. ✓

(3) Low reproductive potential.

✓ (4) Both 1 and 2



74. Exponential growth occurs when there is :-

- ✓ (1) No environmental resistance = unlimited growth
- (2) A fixed carrying capacity
- (3) A great environmental resistance
- (4) resources are limited



75 A species whose distribution is restricted to a small geographical area because of the presence of a competitive superior species is found to expand its distributional range dramatically when the competing species is experimentally removed. This process is known as

- (1) Competitive exclusion principle
- (2) Interference competition
- ✓ (3) Competitive release
- (4) Resource partitioning



76. Thorns as a means of defense is seen in

(1) Acacia ✓

(2) Cactus ✓

✓ (3) Both (1) and (2)

(4) Calotropis



77. Who showed that five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities?

(1) Connell

✓ (2) Mc Arthur

(3) Gause

(4) Darwin



78. Which type of association is found in between entomophilous flower and pollinating agent

- ✓ (1) Mutualism ↩
- (2) Commensalism
- (3) Cooperation
- (4) Co-evolution



79. Parasitism is an interaction between two species in which:

- (1) Both are benefited
- (2) Both are harmed
- (3) One is benefitted and the other is neither benefited nor harmed
- ☒ (4) One is benefited and the other is harmed



80. All organisms living together and interacting with one another in a common environment or natural habitat in a local area are known as:

- (1) Biome
- (2) Biosphere
- (3) Biomass
- ✓ (4) Community



81. Populations evolve to maximise their ____ which is also called as ____

- ✓✓ (1) Reproductive fitness, Darwinian fitness
- (2) Darwinian fitness, sexual fitness
- (3) Verhulst's fitness, Darwinian fitness
- (4) Population fitness, reproductive fitness



V.V.V.V.MP

- ✓(2) Cardiac glycosides

83. Select the correct answer

I. Any species growing exponentially under unlimited resource condition can reach enormous population densities in short time.

II. Darwin showed how even a slow growing animal like tiger could reach enormous numbers in the absence of check. ✗

✓ (1) Only I is correct

(2) Only II is correct

(3) Both are correct

(4) None are correct



84. Which of the following is the best example maintenance of species diversity by predation

- (1) Spreading of Prickly pear cactus in rangeland of Australia ✗
- ✓ (2) Removal of Pisaster fishes in field experiment lead to extinction of more than 10 invertebrate species within a year
- (3) The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body. ✗
- (4) All of these



85. A high density of Elephant population in an area can result in

(1) Predation on one another

(2) Mutualism

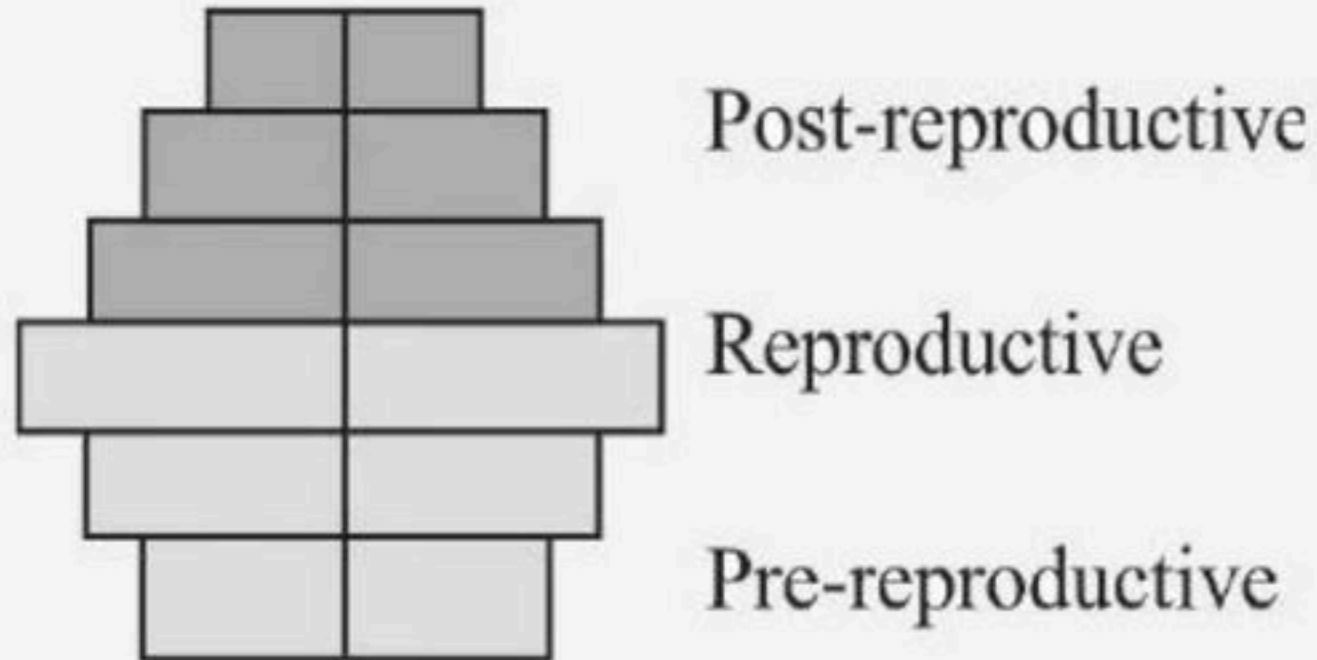
✓ (3) Intraspecific competition ✓

(4) Interspecific competition



86. What type of human population is represented by the following age pyramid?

- (1) Vanishing population
- (2) Stable population
- ✓(3) Declining population
- (4) Expanding population



87. The carrying capacity of a population is determined by its:



- ✓(1) Limiting resources
- (2) Emigration
- (3) Mortality
- (4) Competition



88. 'Competition occurs when closely related species compete for the same resources that are limiting'

Which of the following suggest that the above statement is not entirely true?

(1) Visiting flamingoes and resident fishes ✓✓

✓(2) Resources need not be limiting ✗

(3) Both 1 and 2

(4) None of them



89. How many insects are known to be phytophagous

- ✓ (1) 25% →
- (2) 50%
- (3) 100%
- (4) 18%



90. Which of the following is correct?

- (1) In general, herbivores and plants appear to be more adversely affected by competition than carnivores ✓
- (2) Brood parasitism is seen in cuckoo ✓
- ✓ (3) Both 1 and 2
- (4) None of them



91. Rapid increase in the population is indicated by

- (1) Large number of children
- (2) Large number of young individuals ✓
- (3) Large number of old individuals
- (4) Less number of young individuals



92. When certain exotic species are introduced into a geographical area, they become invasive and start spreading fast because the invaded land does not have its natural ____

- (1) Host
- (2) Parasite
- (3) Prey
- ✓ (4) Predators



93. Who propose competitive exclusion principle?

- (1) Darwin
- (2) Robert May
- ✓ (3) Gause
- (4) Conells



Title: Organism and Population

94 In 2000, for each of the 14 million people present in a country, 0.036 were born and 0.016 died during the year. Using exponential equation, the number of people present in 2010 is predicted as :-

- (1) 14 million
- (2) 2.8 million
- ✓ (3) 16.8 million
- (4) 0.20 million

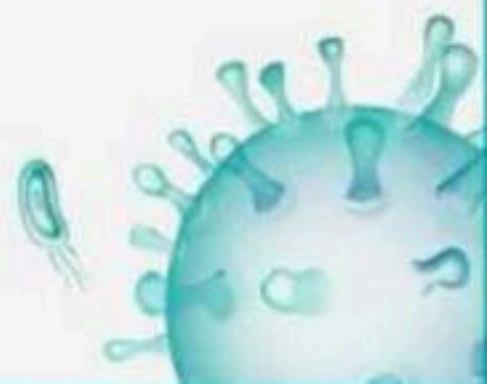
$$\begin{aligned}\frac{dn}{dt} &= rN \\ &= (b-d) \times N \\ &= (0.036 - 0.016) \times 14 \\ &= 0.020 \times 14\end{aligned}$$

$$\frac{dN}{dt} = 0.28$$

$$\frac{dN}{10} = 0.28$$

$$\begin{aligned}dN &= 0.28 \times 10 \\ &= 2.8\end{aligned}$$

$$\begin{aligned}&= 14 + 2.8 \\ &= \underline{16.8}\end{aligned}$$



95. Competition is best defined as a process in which the fitness of one species is significantly how in the presence of another species.

- ☒ (1) Lower
- ☐ (2) Greater
- ☐ (3) Equal
- ☐ (4) None of them



96. Human population shows

- (1) J-shaped growth curve
- (2) Z-shaped growth curve
- ☒ (3) S-shaped growth curve
- (4) All the above



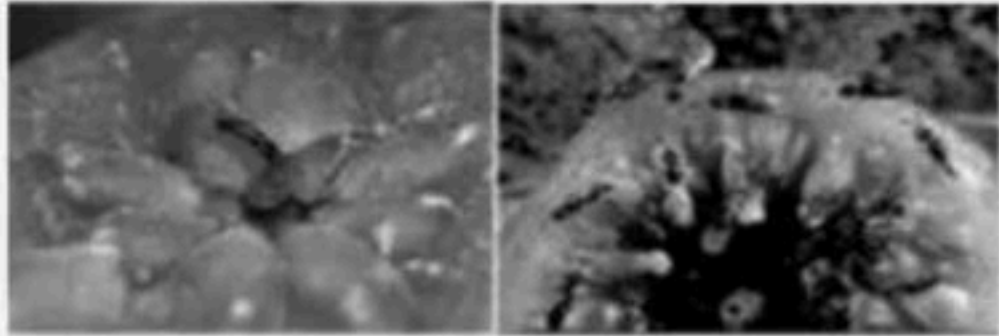
97. If '8' *Drosophila* in a laboratory population of '80' died during a week, the death rate in the population is ____ individuals per *Drosophila* per week

- (1) 10
- (2) 1.0
- (3) zero
- ✓ (4) 0.1

$$d = \frac{8}{80} = 0.1$$



98. The following figure shows



- (1) Fig flower is pollinated by moth
- (2) Orchid flower is pollinated by bee
- (3) Yucca flower is pollinated by wasp
- ✓ (4) Fig flower is pollinated by wasp



99. Select the correct answer. ∴

I. Calotropis produces highly poisonous cardiac glycosidase ✗

II. Visiting flamingoes and resident fishes compete for their common food, the zooplankton ✓

Select the correct answer

(1) Only I is correct

✓ (2) Only II is correct

(3) Both are correct

(4) None are correct



100. Cuscuta is an example of :

- ✓ (1) Ectoparasitism
- (2) Brood parasitism
- (3) Predation
- (4) Endoparasitism



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THANK YOU



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